Mittuniversitetet
MID SWEDEN UNIVERSITY

Försättsblad tentamen / Examination cover

2707183

Anonymitetskod / Anonymous code S - 0020 - ELL		
Kurskod / Course code DT171G	Provkod/ Test code T101	Tentamensdatum / Examination date 2024-03-12
Kursnamn / Course name Datateknik GR (B), Inbäddade system för Internet of Things		
Provnamn / Test name Skriftlig salstentamen		

Skriv din anonymitetskod på varje inlämnat papper
Write your anonymous code on each sheet submitted

Sätt ett kryss (x) för varje inlämnad uppgift
Use an x to indicate which questions has been submitted

Markera nedan med X / Mark below with an X		Poäng / Credit	Lärarens anteckningar / Teacher's notes	Markera nedan med X / Mark below with an X		Poäng / Credit	Lärarens anteckningar / Teacher's notes
1	X	4		16			
2	X	6		17			
3	X	0,5		18			
4	X	2		19			
5	X	4		20			
6	X	2,5		21			
7	X	2,5		22			
8	X	12		23			
9				24			
10				25			
11				26			
12				27			
13				28			
14				29			
15				30			
Poängsumma / Points		33,5	Betyg / Grade C	Lärarsign./ Teachers sign		M.	

Fylls i av tentamensvakt / To be filled in by the invigilator

Antal lösa blad/ No. of sheets submitted 6	**Inlämnad tentamen / Submitted exam** 0	**Leg kontroll / Control identification** ✓	**Sign. tentamensvakt / Sign. invigilator** Lim

2707183

Försättsbladet skall alltid lämnas in även om ingen uppgift behandlats
Examination cover should always be submitted even if no questions are answered

Utskriven 2024-03-08 kl. 09:14:56

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Anonymitetskod/ Anonymus code: S - 0 0 2 0 - E L L

Sidnr / Page no: 1

Lärarens anteckning / Teachers note:

① $i = 25\,mA$, $R_1 = 250\,\Omega$, $R_2 = 150\,\Omega$

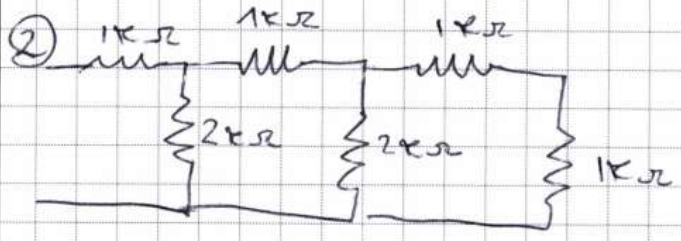
FORMULA:

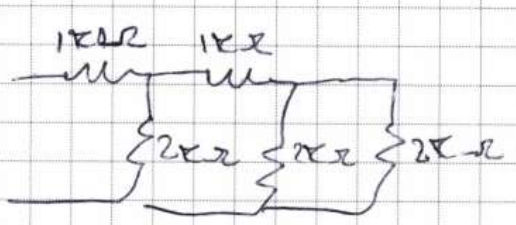
$i = \frac{V}{R} \longrightarrow V = R \cdot i$

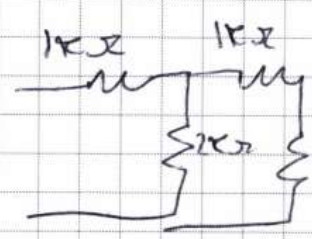
$R_{ep} = R_1 + R_2 = 400\,\Omega$
in series

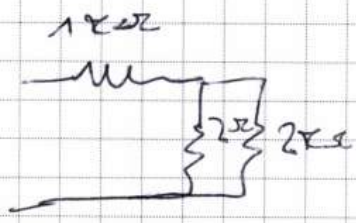
$V = 400 \cdot 25\,mA = 10\,V$

4

② 







$\Rightarrow R_{ep} = 1k + \left(\frac{1}{2k} + \frac{1}{2k}\right)^{-1}$

$= 2k\Omega$

6

Skriv ej på baksidan! Do not write on the other side!

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Anonymitetskod/ Anonymus code: S - 0 0 2 0 - E L L

Sidnr / Page no: 2

Lärarens anteckning / Teachers note:

~~③ $P = \frac{V^2}{R}$ THE NEW RESISTORS HAVE A $P = \frac{1}{4} W = 0,25 W$~~

~~$V = 5V$ $R_{eq} = 66,7 \Omega$~~

~~THE NEW R_{eq} SHOULD BE~~

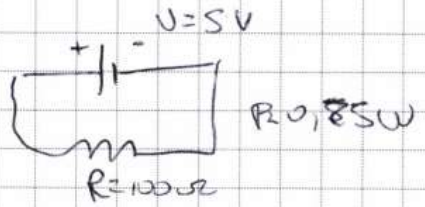
~~$R = \frac{V^2}{P} = 100 \Omega$~~

③ IN ORDER TO HAVE A NEW CIRCUIT WITH A F.E.M. OF 5V AND POWER 0,25 W, WE NEED A RESISTOR WITH THIS PARAMETER:

0,5

$P = \frac{V^2}{R}$, THE NEW EQUIVALENT RESISTANCE IS:

$$R_{eq} = \frac{V^2}{P} = \frac{5^2}{0,25} = 100 \Omega$$ ✓



Skriv ej på baksidan! Do not write on the other side!

Lärarens anteckning / Teachers note:

5) PWM STANDS FOR PULSE WIDTH MODULATION. IT IS A WAY OF REPRESENTING A SIGNAL WITH RECTANGLE DIGITAL AT VARYING DUTY CYCLES. IT IS A TECNIQUE TO OBTAIN ANALOG RESULTS WITH DIGITAL MEANS.

4

e.g.

6) THE DIFFERENCE BETWEEN CONNECTING DIGITAL OR ANALOG SENSORS TO A MICROCONTROLLER ARE THE TYPE OF CONVERTERS THAT WILL BE USED IN THE CASE OF A DIGITAL SENSOR IT WILL BE USED THE DAC: DIGITAL TO ANALOG CONVERTER, WHICH TAKES IN INPUT A DIGITAL SIGNAL AND CONVERTS IT INTO AN ANALOG ONE, FOR EXAMPLE AN AUDIO CARD OF A COMPUTER.

2,5

ON THE OTHER HAND THERE IS THE ADC: ANALOG TO DIGITAL CONVERTER, WHICH TAKES IN INPUT AN ANALOG SIGNAL AND CONVERTS IT INTO A DIGITAL ONE, FOR EXAMPLE A SAMPLER OR A TELEPHONE MODEM.

Lärarens anteckning / Teachers note:

(7) MQTT IS AN APPLICATION PROTOCOL NORMALLY AVAILABLE AT PORT 1883 OF THE STACK TCP/IP. IT IS A COMMUNICATION PROTOCOL THAT THROUGHT CLIENTS NAMED "SUBSCRIBERS" AND "PUBLISHERS" SEND DATA ON THE NETWORK.
THE ACTUAL TRASFER OF DATA HAPPENS WITH A SERVER, CALLED "BROKER".
THE ~~SUB~~ CLIENTS SUBSCRIBE TO A CERTAIN TOPIC ~~AND~~ FROM THE BROKER, AND PUBLISHERS SEND MESSAGES TO THE BROKER ON A SPECIFIC TOPIC THAT WILL BE DELIVERED TO THE SUBSCRIBERS.
NORMALLY THE OLD MESSAGES ARE NOT STORED ON SERVER, SO A NEW SUBSCRIBER WON'T BE ABLE TO SEE OLD MESSAGES. (CLIENT $\xrightarrow{\text{PUBLISH}}$ BROKER $\xleftarrow{\text{SUBSCRIBE}}$ CLIENT)

Security Authentication

2,5

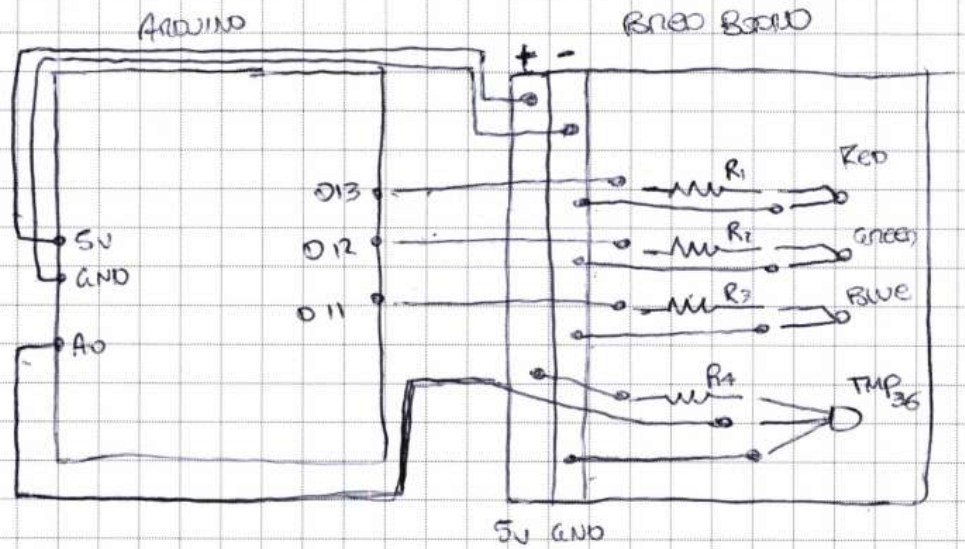
(4) AN OPTOCOUPLER IS A SEMICONDUCTOR DEVICE WHICH ENABLES TO CHANGE THE VOLTAGE ON AN ELECTRICAL CONNECTION. A USAGE OF THIS DEVICE CAN BE TO MAKE A CONNECTION ~~BETW~~ BETWEEN TWO DEVICES WHICH WORKS ON DIFFERENT ~~VOLTA~~ TENSIONS, FOR INSTANCE AN ARDUINO (5V) AND A RASPBERRY (3.3V)

2

Galvanic isolation?

Lärarens anteckning / Teachers note:

(8)



CIRCUIT DIAGRAM

COMPONENTS USED:

1 BREADBOARD

1 ARDUINO UNO

3 LEDs (RED, GREEN, BLUE)

1 TMP36 (TEMPERATURE SENSOR)

10 CABLES

4 RESISTORS (R4 = ~660KΩ TO MAKE A BIT LESS WATTAGE, AND R1 = R2 = R3 = 220KΩ, IT SHOULD BE A GOOD AMOUNT FOR THE LEDS)

CODE:

```
VOID SETUP() {        //SET PINS, POWER OFF LEDS
  PINMODE(13, OUTPUT);
  PINMODE(12, OUTPUT);
  PINMODE(11, OUTPUT);
```

Lärarens anteckning / Teachers note:

```
  DIGITALWRITE(13, LOW);
  DIGITALWRITE(12, LOW);
  DIGITALWRITE(11, LOW);
  PINMODE(A0, INPUT);
}
VOID LOOP(){                    //ADC CONVERTS SIGNAL TO
                                // DIGITAL AND CALCULATE THE
                                //TEMPERATURE
  INT SENSORPIN = A0;
  INT READING = ANALOGREAD(SENSORPIN);
  FLOAT VOLTAGE = READING * (5,0 / 1024.0);
  FLOAT TEMPERATUREC = (VOLTAGE * 0,5) * 100;
IF(TEMPERATUREC > 30){          //LIGHT RED LED
  DIGITALWRITE(13, HIGH);
  DIGITALWRITE(12, LOW);
  DIGITALWRITE(11, LOW);
}
ELSE IF (TEMPERATUREC < 15){    //LIGHT BLUE LED
  DIGITALWRITE(13, LOW);
  DIGITALWRITE(12, LOW);
  DIGITALWRITE(11, HIGH);
}
ELSE{                           // LIGHT GREEN LED
  DIGITALWRITE(13, LOW);
  DIGITALWRITE(12, HIGH);
  DIGITALWRITE(11, LOW);
}
DELAY(5000);  // CHECK EVERY 5SECONDS
}
```

12

Nok